

Clinical Monitoring & Intelligent Rehabilitation Platform

AI-powered clinical decision support for rehabilitation monitoring

Team: [NEXORA]

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CHARUSAT

MONITOR • DETECT • ASSIST



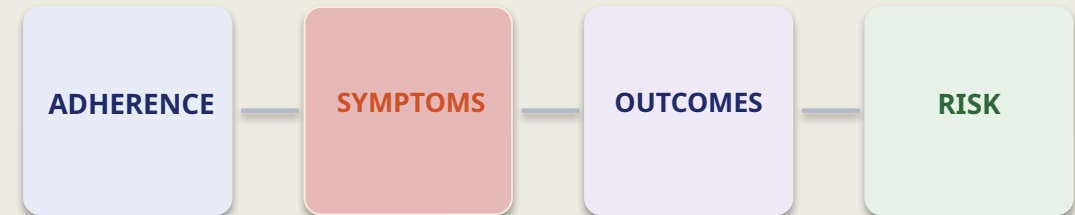
Venous Rehabilitation Has a Monitoring Gap

The challenge is not only delivering rehabilitation — it is knowing when the journey is changing.

THE CHALLENGE

- Patients with venous conditions can experience pain, heaviness, swelling, fatigue and reduced quality of life.
- Prolonged standing and impaired calf-muscle pumping can contribute to venous symptoms.
- Clinicians need visibility into rehabilitation adherence, symptom/outcome trends and session patterns over time.
- A monitoring workflow should help surface meaningful changes for clinician review.

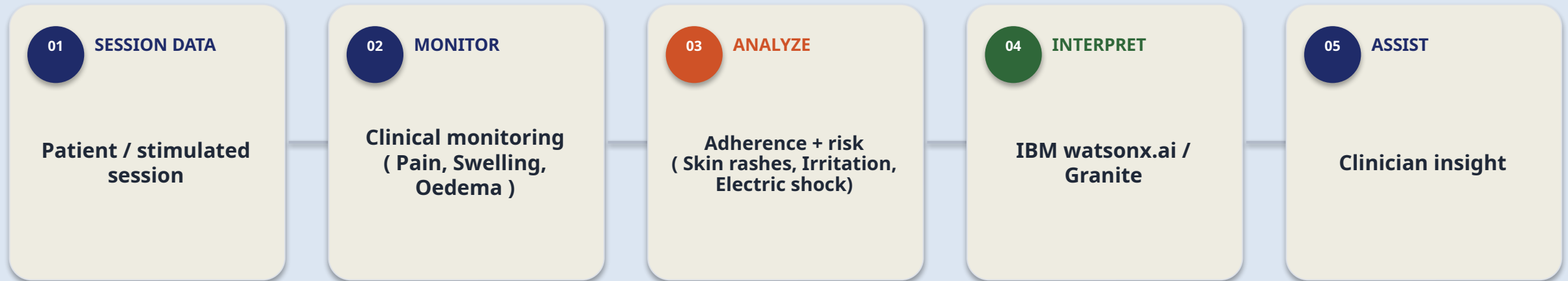
WHAT NEEDS TO BE VISIBLE



Monitoring turns scattered session information into a reviewable clinical picture.

From Session Data to Clinical Insight

A digital monitoring layer that connects rehabilitation data, deterministic analysis and grounded AI.



DECISION SUPPORT — NOT DIAGNOSIS

Veno-Pump helps clinicians track sessions, identify adherence/risk patterns and generate grounded AI-assisted clinical insights.

A Complete Digital Monitoring Workflow

One platform for monitoring → analysis → insight → clinical review.

01

PATIENT MANAGEMENT

Profiles and rehabilitation history.

02

SESSION MONITORING

Simulate/record sessions and patient feedback.

03

ADHERENCE TRACKING

Planned vs. completed sessions and trends.

04

RISK & ALERTS

Rule-based detection of meaningful changes in adherence, pain and session consistency.

05

AI CLINICAL INSIGHTS

Patient-specific summaries grounded in backend-calculated metrics.

06

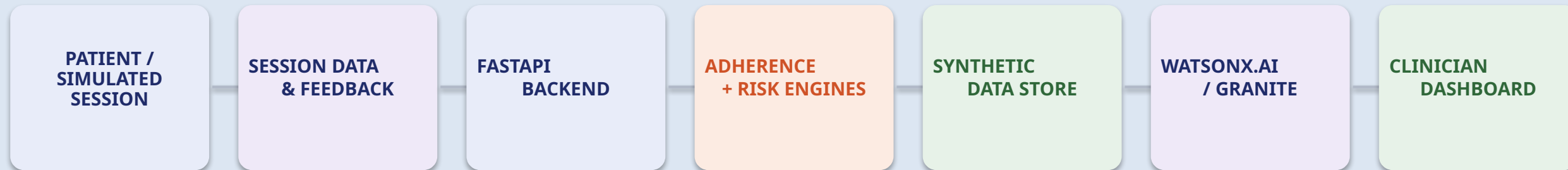
AI ASSISTANT & REPORTS

Conversational support plus structured/plain-text reporting.

Implemented focus: clinical monitoring prototype + synthetic/simulated rehabilitation data.

Clinical Monitoring Workflow

Deterministic monitoring first; grounded generative AI second.



GROUNDING RULE

The AI receives authoritative backend-calculated metrics rather than inventing or independently recalculating them.

Deterministic monitoring + grounded generative AI = clinician-focused decision support

Technical Architecture & Deployment

A modular web stack with a dedicated AI service and transparent monitoring engines.

ARCHITECTURE FLOW

FRONTEND React + TypeScript + Vite + Tailwind + Recharts

BACKEND Fast API + SQL Model | REST APIs

DATA SQLite | synthetic / demo patient data

MONITORING Adherence engine + risk engine + AI service

AI IBM watsonx.ai | Granite 4 H Small

DEPLOYMENT

 **VERCEL**
Frontend

 **RENDER**
Backend

 **IBM CLOUD**
watsonx.ai

LIVE DEMO

veno-pump.vercel.app

Backend: veno-pump-api.onrender.com

IBM-Powered Intelligence & Development

Two distinct roles: Bob accelerates development; watsonx.ai + Granite powers runtime intelligence.



IBM BOB

DEVELOPMENT PARTNER

- Used as the AI-powered software development partner during the project.
- Supported planning, implementation, debugging, testing/review and refinement.
- Development-time contributor — not a runtime component of the deployed app.



watsonx.ai + Granite

RUNTIME AI CAPABILITY

- Integrated into the deployed backend AI service.
- Granite 4 H Small generates patient-specific clinical insights from grounded platform data.
- Uses authoritative adherence, session, outcome and risk metrics.

"IBM Bob accelerated our development lifecycle, while IBM watsonx.ai provides the runtime generative AI capability."

From Patient Data to Clinical Insight

Working demo journey — use real application captures during the final submission demo.

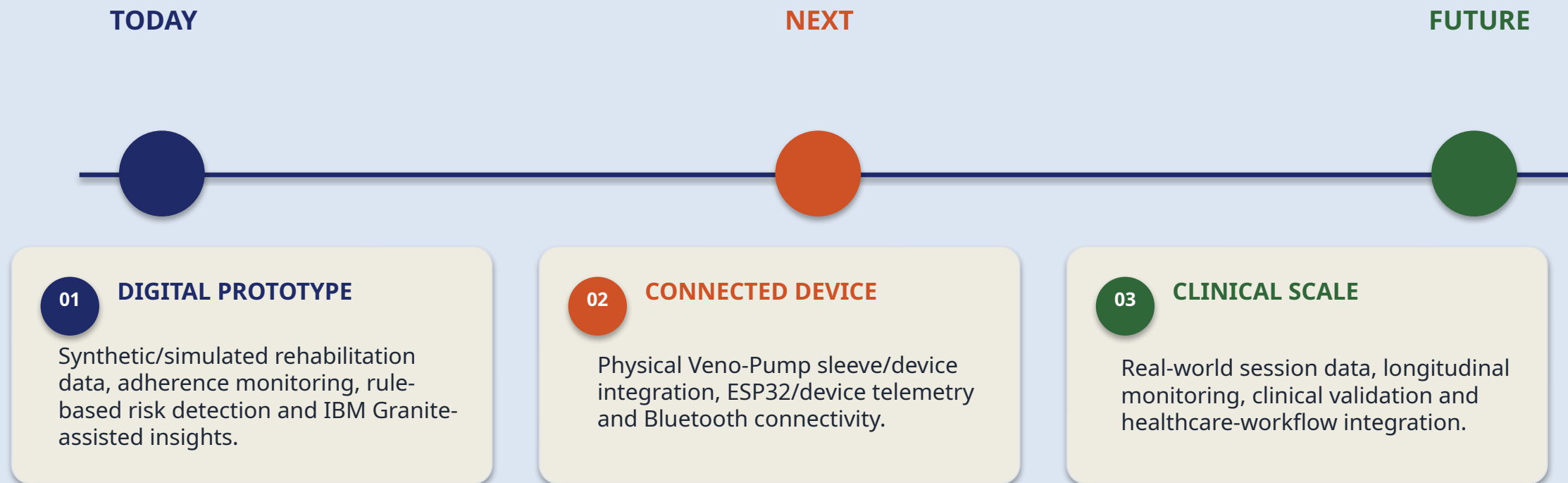


OPEN APP → DASHBOARD → P-004 → ADHERENCE / OUTCOMES / RISK → GRANITE INSIGHT

Synthetic demo data • Real backend • Real IBM AI output

Beyond the Hackathon

Today: Digital Intelligence Layer → Future: Connected Veno-Pump Ecosystem



Device integration is a future development phase — not a current platform capability.

Making Rehabilitation Monitoring Intelligent

Veno-Pump connects rehabilitation data, deterministic monitoring and IBM AI to help clinicians identify meaningful changes and make better-informed decisions.

MONITOR

Track rehabilitation journeys and adherence.

DETECT

Identify meaningful changes in symptoms, outcomes and risk patterns.

ASSIST

Generate grounded AI insights for clinician review.

LIVE DEMO

veno-pump.vercel.app/

GitHub • [Krishna5455/bob-ai-hackathon-NEXORA](https://github.com/Krishna5455/bob-ai-hackathon-NEXORA)